

A Framework for Decentralized Governance and Automated Threat Mitigation in Agentic AI Infrastructures

Kalyan K. Parameshwaran
Massachusetts Institute of Technology (MIT)
kparam@mit.edu
Charles E. Leiserson
Massachusetts Institute of Technology (MIT)
cel@mit.edu

March 16, 2023

Abstract

The rapid transition from static, single-turn Large Language Model (LLM) deployments to autonomous, multi-agent networks has outpaced the conceptual and technical apparatus of contemporary information security. Agentic AI systems, capable of dynamic goal decomposition, independent tool-use, and unsupervised interaction with critical enterprise infrastructure, introduce a class of non-deterministic vulnerabilities that classical, rule-based security architectures were never designed to contain. This research develops and evaluates an integrated socio-technical framework for governing agentic infrastructures, uniting three previously disconnected theoretical traditions: Dorst’s Problem-Solution Co-Evolution model from Design Science, an Algorithmic Zero-Trust extension of cybersecurity architecture, and Boundary Objects Theory drawn from innovation and process management. At the technical core of the framework is the Security Orchestration Gateway (SOG), a semantic abstraction layer that intercepts agent-to-agent and agent-to-system communication, compiles probabilistic intent into a deterministic evaluation matrix, and enforces compliance against Common Criteria EAL 4+ policy bounds. The framework is evaluated across three methodological phases: (I) artifact design of the SOG; (II) empirical, adversarial simulation within a sandboxed enterprise environment using quantitative metrics for defensive efficiency, latency overhead, and throughput degradation; and (III) an organizational boundary-spanning analysis using Social Network Analysis (SNA) to examine how cross-functional teams coordinate during live security incidents. The research contributes a computationally codified extension of co-evolutionary design theory, a validated methodology for extending Zero-Trust principles to non-deterministic agent intent, and an implementation-ready toolkit for enterprise leaders seeking to reconcile innovation velocity with systemic risk mitigation in regulated, safety-critical industries.

Keywords: agentic AI, zero-trust architecture, socio-technical systems, boundary objects, security orchestration, algorithmic governance, resilience engineering

Contents

1	Introduction and Research Problem	3
1.1	Background and Motivation	3
1.2	The Paradigm Shift to Agentic AI	3
1.3	The Security and Operational Bottleneck	3
1.4	Research Questions	4
1.5	Significance of the Study	4
1.6	Structure of the Remainder of this Document	5
2	Theoretical Framework	5
2.1	Systems and Design Science: Problem-Solution Co-Evolution	5
2.2	Cybersecurity Architecture: Algorithmic Zero-Trust and Behavioural Telemetry	5
2.3	Innovation and Process Management: Boundary Objects and Structural Resilience	6
2.4	Integrated Conceptual Model	6
3	Literature Review	7
3.1	Vulnerabilities in Autonomous LLM Architectures	7
3.2	The Limitations of Modern Enterprise Security Frameworks	7
3.3	Socio-Technical System Resilience	7
3.4	Synthesis and Identified Gap	8
4	Methodology and System Architecture	8
4.1	Phase I: Framework Artifact and “Security Orchestration Gateway” Design	8
4.2	Phase II: Empirical Simulation, Adversarial Testing, and Quantitative Analytics	9
4.2.1	Quantitative Evaluation Metrics	9
4.3	Phase III: Organizational Boundary Spanning Analysis	10
4.4	Validity, Reliability, and Ethical Considerations	10
5	Anticipated Contributions, Discussion, and Industrial Impact	10
5.1	Theoretical and Academic Contributions	10
5.2	Industrial and Practical Impact	11
5.3	Discussion	11
6	Limitations	11
7	Future Work	11
8	Conclusion	12
A	Glossary of Terms	13

1 Introduction and Research Problem

1.1 Background and Motivation

Over the past decade, enterprise adoption of artificial intelligence has proceeded through several distinct architectural generations. The earliest generation consisted of narrow, task-specific machine learning models embedded within otherwise conventional software pipelines. The second generation, catalyzed by the emergence of large-scale transformer-based language models, introduced conversational and generative capabilities but remained fundamentally reactive: a human initiated each interaction, and the model’s influence over external systems was mediated through tightly scoped, human-reviewed interfaces. The current, third generation, commonly referred to as Agentic AI, breaks decisively from this reactive posture. Agentic systems plan, decompose, and execute multi-step workflows with minimal human mediation, invoking external tools, application programming interfaces (APIs), and infrastructure-as-code pipelines as a matter of routine operation.

This capability shift is not merely incremental. It represents a change in the underlying trust model of enterprise computing. Where earlier generations of software behaved deterministically, the same input reliably producing the same output, agentic systems behave probabilistically, and their behavior is additionally conditioned on an evolving, self-modifying set of instructions, plans, and intermediate reasoning traces. The security, compliance, and organizational structures that enterprises have spent decades refining were built for the deterministic world. This research is motivated by the widening gap between what agentic infrastructures are capable of doing and what existing governance frameworks are capable of safely permitting.

1.2 The Paradigm Shift to Agentic AI

The architectural landscape of Artificial Intelligence is undergoing a fundamental paradigm shift. The industry is rapidly transitioning from static, human-initiated Large Language Model (LLM) implementations, such as single-turn chatbots and isolated Retrieval-Augmented Generation (RAG) frameworks, to autonomous, multi-agent networks.

These emerging Agentic AI ecosystems are characterized by their capacity for open-ended execution, dynamic goal decomposition, and independent tool-use. In an enterprise context, these agents are granted the authority to execute complex workflows, orchestrate software-defined infrastructure, mutate critical databases, and interface with external application programming interfaces (APIs) with little to no human oversight. Multi-agent orchestration frameworks now routinely chain together dozens of discrete tool invocations per task, with individual agents delegating sub-goals to other agents in a recursive fashion. The resulting execution graphs are frequently opaque even to the engineers who designed the originating system, since the precise sequence of tool calls is determined at runtime by the model’s own probabilistic reasoning rather than by a fixed, auditable control-flow graph.

1.3 The Security and Operational Bottleneck

While Agentic AI promises exponential leaps in operational efficiency, it introduces unprecedented, non-deterministic vulnerabilities that break traditional software security models. Classical information security paradigms rely on predictable state-machine behaviors, rigid access control lists (ACLs), perimeter defenses, and signature-based intrusion detection systems. Agentic networks, by contrast, operate probabilistically, and a security control that is effective against one execution trace may be entirely blind to a semantically equivalent but syntactically novel attack path generated moments later.

This introduces critical vectors such as:

- **Cascading Algorithmic Failures:** Small perturbations in one agent’s output compounding into systemic operational shutdowns across an entire network. Because agentic systems are frequently composed into pipelines in which one agent’s output becomes another agent’s input, a single hallucinated instruction or subtly corrupted intermediate artifact can propagate downstream with amplifying effect, analogous to error propagation in tightly coupled mechanical systems.
- **Prompt Injection and Semantic Hijacking:** Adversarial inputs altering an agent’s core instruction set, causing it to abuse its access privileges (e.g., exfiltrating data or wiping cloud storage). Indirect variants of this attack are especially concerning, as the malicious instruction need not originate from the nominal user at all, but may be embedded within a document, webpage, or API response that the agent ingests as ordinary data.
- **The Innovation-Security Conflict:** A stark operational bottleneck where strict, legacy compliance frameworks (e.g., Common Criteria EAL 4+ standards) stifle the deployment velocity and adaptability of AI networks, while unmitigated agent autonomy exposes the enterprise to existential security risks. Enterprises are consequently caught between two failure modes: over-constraining agents to the point of eliminating their economic value, or under-constraining them to the point of unacceptable exposure.

1.4 Research Questions

This paper examines how enterprises can defend against the security threats posed by autonomous AI agents, not through episodic remediation, but through a sustained, adaptive governance approach. This problem became apparent when it was observed that existing security models, designed for human users and static services, were not constructed to accommodate software capable of independent reasoning, decision making, and action.

This observation informed the direction of the research in three respects. First, rather than treating agentic threats as a fixed target subject to periodic patching, the research sought to understand how the threat landscape and the solutions developed to address it could be modeled as a system that evolves jointly, in which each dimension continuously adapts in response to the other.

Second, it was recognized that Zero Trust security, in its original conception, was developed to verify identity, confirming that a person or service is who it claims to be. Autonomous agents, however, do not merely authenticate; they act with intent. This raised the question of what technical architecture would be required to extend Zero Trust principles beyond identity verification, such that the meaning and intent underlying an agent’s behavior could be evaluated directly.

Third, the research sought to determine whether organizational structure itself influences security outcomes, specifically, whether cross-functional teams, coordinated through shared reference artifacts or “boundary objects,” demonstrate greater capacity to detect, interpret, and respond to agentic security incidents than conventional, siloed engineering teams.

1.5 Significance of the Study

The significance of this work is twofold. Academically, it extends design science, cybersecurity architecture, and innovation management theory into a domain, autonomous algorithmic infrastructure, that has so far been addressed by these literatures only in isolation. Practically, it offers technology leaders in regulated industries a concrete, implementation-ready pathway for deploying agentic systems without abandoning the compliance postures required by law, regulation, or institutional risk tolerance.

1.6 Structure of the Remainder of this Document

Section 2 develops the theoretical framework synthesizing Design Science, cybersecurity architecture, and innovation management. Section 3 situates the work within the existing literature on LLM vulnerabilities, enterprise security frameworks, and socio-technical resilience. Section 4 details the three-phase research methodology, including the design of the Security Orchestration Gateway and the quantitative metrics used to evaluate it. Section 5 discusses anticipated contributions and industrial impact. Section 6 addresses the limitations of the proposed approach, and Section 7 concludes.

2 Theoretical Framework

To construct a comprehensive solution to this socio-technical challenge, this research synthesizes three distinct theoretical domains, building a unified paradigm for algorithmic infrastructure governance. No single existing discipline is sufficient on its own: design science offers a model of how problems and solutions evolve together but says little about cybersecurity mechanics; Zero-Trust architecture offers rigorous authentication and authorization mechanics but is agnostic to organizational coordination; and innovation management offers a theory of cross-functional coordination but was not developed with autonomous, non-human actors in mind. The framework’s central theoretical claim is that these three domains are not merely complementary but structurally interdependent when applied to agentic infrastructure.

2.1 Systems and Design Science: Problem-Solution Co-Evolution

This thesis rejects the notion that security threats are static variables to be patched post-deployment. Instead, it adapts Kees Dorst’s Problem–Solution Co-evolution Model from Design Science (Dorst, 2019). In the original formulation, designers do not first fully specify a problem and then search for a solution; rather, the act of proposing candidate solutions reshapes the designer’s understanding of the problem itself, and the two co-evolve through iterative cycles.

In complex agentic infrastructures, the threat landscape (problem space) changes dynamically as agents mutate their own workflows (solution space). Every new capability granted to an agent, a new tool, a new data source, a new downstream system it may write to, simultaneously expands the attack surface and reshapes what a “normal” baseline of behavior looks like. This framework models security threat modelling and system capability deployment as an intertwined, evolving pair, ensuring that defence mechanisms adapt at the exact velocity of agentic iteration, rather than lagging behind it in a conventional patch-and-release cycle.

A further implication of adopting the co-evolutionary lens is that the framework treats “design fixation”, the tendency of designers to over-commit to an early solution concept and resist revising it in light of new problem information (Jansson & Smith, 1991), as a specific organizational risk in agentic governance. Security teams that fixate on a fixed taxonomy of known attack patterns are structurally prone to missing the emergent, self-modifying threat classes unique to agentic systems.

2.2 Cybersecurity Architecture: Algorithmic Zero-Trust and Behavioural Telemetry

The core technical layer expands upon contemporary Zero-Trust Architectures (ZTA). While classical ZTA authenticates human users and static services, this framework develops an Algorithmic Zero-Trust model. It integrates concepts from Network Detection and Response (NDR) and Endpoint Detection and Response (EDR), repurposing them for language-model payloads. Rather than inspecting packets for known malware signatures, it analyzes the semantic telemetry and intent streams of agent-to-agent communication to establish behavioral baselines and isolate anomalous intents.

Table 1: Theoretical domains synthesized in the proposed framework.

Theoretical main	Do-	Core Methodology / Model		Operational Application in Agentic AI Systems
Systems & Design Science		Dorst’s Solution Evolution Model	Problem–Co-	Rejects static security patching; treats the threat landscape (problem) and agent workflows (solution) as an intertwined, dynamically evolving pair.
Cybersecurity Architecture	Ar-	Algorithmic Trust & Telemetry	Zero- Behavioural	Shifts from human/static service authentication to analysing semantic intent streams and agent-to-agent communication baselines (NDR/EDR for language payloads).
Innovation & Process Management	Pro-	Lean Six Sigma & Boundary Theory	Sigma & Objects	Uses cryptographic logs, compliance matrices, and dashboards as “boundary objects” to let engineers, compliance officers, and auditors co-manage the system without dropping velocity.

Three design principles distinguish Algorithmic Zero-Trust from its classical antecedent. First, the unit of authentication is not an identity but an intent, a structured representation of what an agent is attempting to accomplish, extracted from its natural-language reasoning trace. Second, trust is never granted persistently; each discrete action is re-evaluated against the current policy and behavioral baseline, regardless of whether the immediately preceding action was approved. Third, anomaly detection operates at the semantic layer rather than the syntactic layer, meaning that two superficially different tool-call sequences that achieve the same underlying (and impermissible) goal are treated as equivalent threats.

2.3 Innovation and Process Management: Boundary Objects and Structural Resilience

Because an enterprise infrastructure is inherently socio-technical, the framework incorporates Boundary Objects Theory (Star & Griesemer, 1989) alongside Lean Six Sigma methodologies (George, 2002). Boundary objects, in Star and Griesemer’s original formulation, are artifacts that are simultaneously flexible enough to be locally meaningful within distinct communities of practice, yet robust enough to maintain a common identity across those communities.

Cryptographic validation logs, automated compliance matrices, and real-time observability dashboards are treated as boundary objects. These artifacts allow distinct, often conflicting organizational groups, such as pure software engineers, corporate compliance officers, and external regulatory auditors, to safely co-manage and optimize the system without sacrificing operational throughput or security integrity. Lean Six Sigma contributes a complementary process-optimization vocabulary: the framework treats unnecessary security friction (redundant approvals, duplicated telemetry review, unclear escalation paths) as a form of process “waste” to be systematically identified and eliminated, without weakening the underlying control itself.

2.4 Integrated Conceptual Model

Bringing these three domains together, the framework can be summarized as a closed-loop governance cycle: (1) the co-evolutionary lens continuously reframes what counts as the “problem” as agent capabilities expand; (2) the Algorithmic Zero-Trust layer operationalizes real-time detection and containment of

impermissible intent; and (3) boundary objects translate the technical telemetry generated by that containment layer into artifacts that distinct organizational stakeholders can jointly interpret and act upon. None of the three layers is sufficient in isolation: a purely technical Zero-Trust layer without boundary objects produces telemetry that compliance and audit functions cannot act on, while boundary objects without a co-evolutionary problem model quickly become stale as agent capabilities change.

3 Literature Review

3.1 Vulnerabilities in Autonomous LLM Architectures

Early research into LLM vulnerabilities focused heavily on static adversarial prompts designed to bypass safety filters (Zou et al., 2023). These early attacks were largely characterized as a single-turn phenomenon: an adversary crafts a specific input string, submits it once, and either succeeds or fails in eliciting prohibited output. However, as systems have evolved into multi-agent frameworks utilizing tool-use and RAG architectures (Chase, 2023), the attack surface has expanded dramatically, both in the number of distinct entry points available to an adversary and in the temporal persistence of a successful compromise.

Recent literature demonstrates that agents can be manipulated via indirect prompt injections embedded within third-party data streams (Greshake et al., 2023), turning autonomous agents into vectors for remote code execution and unauthorized data extraction. The indirect injection class of attack is particularly consequential for enterprise deployment, because it does not require the adversary to have any direct access to the agent’s interface at all; it is sufficient for the adversary to control content that the agent will, in the ordinary course of its assigned task, later retrieve and process. This shifts the security perimeter from the point of human interaction to the entire universe of external content an agent might ingest, a universe that is, for most enterprise deployments, effectively unbounded.

3.2 The Limitations of Modern Enterprise Security Frameworks

Contemporary enterprise security frameworks, notably NIST’s Zero Trust Architecture (Rose et al., 2020), operate on the maxim of “never trust, always verify.” While highly effective for deterministic access patterns, these models struggle with the fluid API calls and dynamic code generation inherent to agentic pipelines. Security tooling such as traditional Web Application Firewalls (WAFs) fail to parse semantic manipulation, leaving a structural gap in identifying when an authorized agent is executing an unauthorized intent.

This gap is best understood as a mismatch of representational granularity. Classical ZTA verifies discrete, well-defined actions, a specific API call, a specific file access, against a comparatively static policy. Agentic systems, however, compose primitive actions into emergent higher-order behaviors that may not map cleanly onto any single policy rule, even when every individual action in the sequence is, considered alone, entirely permissible. An agent that is authorized to read customer records and separately authorized to send outbound email is not thereby authorized to exfiltrate customer data via email, yet a classical ACL-based system has no native mechanism for recognizing the composite action as distinct from its permissible constituent parts.

3.3 Socio-Technical System Resilience

In the realm of operations and innovation management, resilience is defined as a system’s capacity to absorb disturbance while maintaining its core functional structure (Holling, 1973; Perrow, 1984). The integration of highly automated processes within complex organizations frequently introduces “normal accidents” due to interactive complexity and tight coupling, a diagnosis that Perrow originally developed for physical

infrastructure such as nuclear power plants and chemical processing facilities, but which maps with striking precision onto multi-agent AI pipelines, which likewise exhibit both high interactive complexity (many components can affect many other components in non-obvious ways) and tight coupling (failures propagate quickly, with little slack time for human intervention).

Lean Six Sigma tools have historically been utilized to decouple these processes and eliminate operational waste (George, 2002). However, literature regarding the application of these process optimization frameworks to the governance of non-deterministic, autonomous algorithmic structures remains scarce. This thesis directly addresses this vacuum by treating the coordination overhead between security, compliance, and engineering functions as a process subject to the same continuous-improvement discipline traditionally applied to manufacturing and service delivery workflows.

3.4 Synthesis and Identified Gap

Taken together, the literature reviewed above establishes three separate findings: that agentic systems are vulnerable to an expanding class of semantic and indirect attacks; that existing enterprise security architectures are structurally mismatched to the granularity at which agentic systems actually operate; and that organizational resilience theory, while directly applicable in principle, has not yet been operationalized for autonomous, non-human infrastructure. No existing study integrates all three strands into a single, empirically testable governance framework. This gap motivates the methodology developed in Section 4.

4 Methodology and System Architecture

This research utilizes an engineering systems design approach executed across three sequential phases, moving from artifact design to empirical validation, and finally to organizational analysis. The overall research design follows a design-science research paradigm: theoretical constructs are first instantiated as a concrete technical artifact (the Security Orchestration Gateway), the artifact is then subjected to controlled empirical stress-testing, and the resulting insights are triangulated against qualitative organizational data.

4.1 Phase I: Framework Artifact and “Security Orchestration Gateway” Design

The initial phase focuses on the technical architecture of the guardrails. A specialized semantic abstraction layer, termed the Security Orchestration Gateway (SOG), is engineered to intercept all agent-to-agent and agent-to-system communications. The SOG operates by breaking down agent payloads using a custom-developed semantic compiler. It translates probabilistic language models into a structured, deterministic evaluation matrix. This matrix checks agent intents against predefined enterprise policies and cryptographic bounds required for Common Criteria EAL 4+ compliance.

The semantic compiler component performs three sequential transformations on an incoming agent payload. First, it parses the agent’s stated reasoning trace and proposed tool invocation into a normalized intermediate representation, stripping away superficial phrasing variance while preserving the underlying goal structure. Second, it cross-references this normalized representation against a continuously updated behavioral baseline for the originating agent, flagging deviations that exceed a configurable statistical threshold. Third, it emits a structured evaluation record, the deterministic evaluation matrix, that downstream policy engines can consume without themselves needing to reason probabilistically.

This architecture is deliberately designed so that the boundary between the probabilistic agentic layer and the deterministic enforcement layer is drawn as early as possible in the request path. All auditable, compliance-relevant decisions occur strictly within the deterministic evaluation matrix, which can be logged, replayed, and certified against EAL 4+ evidentiary requirements independent of the underlying model’s non-reproducible internal reasoning.

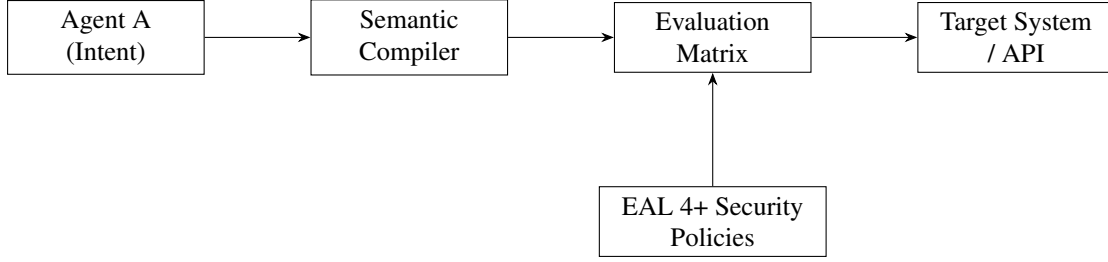


Figure 1: Security Orchestration Gateway (SOG) request path. Agent intent is normalized by the Semantic Compiler, checked by the Evaluation Matrix against EAL 4+ security policies, and, if approved, forwarded to the target system or API.

4.2 Phase II: Empirical Simulation, Adversarial Testing, and Quantitative Analytics

To evaluate the resilience of the SOG under extreme duress, a high-fidelity, sandboxed enterprise infrastructure is constructed. This environment hosts a multi-agent network managing simulated corporate data pipelines, asset allocations, and API-driven workflows, mirroring the operational profile of a mid-to-large regulated enterprise.

An automated adversarial evaluation engine is introduced to simulate ongoing red-team attacks. The network is subjected to continuous strains, including semantic data poisoning, indirect prompt injections, and multi-vector API hijacking. The adversarial engine is designed to escalate in sophistication across successive simulation rounds, beginning with previously catalogued attack patterns drawn from the literature reviewed in Section 3 and progressing toward novel, combinatorial attack sequences generated specifically to probe the boundaries of the SOG’s behavioral baseline.

4.2.1 Quantitative Evaluation Metrics

Systemic resilience and performance trade-offs are explicitly calculated using the following mathematical formulations.

Defensive Efficiency (E_d): Measures the ratio of successfully contained adversarial exploits (C_a) against the total number of injected exploits (I_a):

$$E_d = \frac{C_a}{I_a}. \quad (1)$$

System Latency Overhead (ΔL): Calculates the processing delay introduced by the SOG’s semantic compilation, where L_g represents total gated system latency and L_b represents unshielded baseline latency:

$$\Delta L = L_g - L_b. \quad (2)$$

Throughput Degradation Coefficient (D_t): Evaluates the operational penalty of security friction, comparing gated transaction throughput (T_g) against ideal baseline throughput (T_b):

$$D_t = 1 - \frac{T_g}{T_b}. \quad (3)$$

By mapping the interaction between E_d and D_t across different security profiles, this research uncovers the optimal mathematical frontier where system security does not catastrophically degrade process efficiency. Each of the three metrics is computed at multiple SOG configuration settings, ranging from a

permissive profile that prioritizes throughput to a maximally restrictive profile that prioritizes containment, so that the resulting frontier can be presented as a continuous trade-off curve rather than a single operating point. This allows enterprise adopters to select a configuration appropriate to their specific regulatory and risk-tolerance context rather than inheriting a one-size-fits-all default.

4.3 Phase III: Organizational Boundary Spanning Analysis

The final phase addresses the human element of the infrastructure. Utilizing a multiple-case study design, the research tracks human operators during active security incident simulations.

Four highly cross-functional, cognitively diverse engineering teams (comprising AI developers, cybersecurity specialists, and compliance personnel) are contrasted against four homogeneous software engineering teams. The two team configurations are given identical incident scenarios, identical access to the SOG's telemetry dashboards, and identical time budgets, so that any observed difference in detection speed, remediation quality, or communication pattern can be attributed to team composition rather than to differences in the underlying incident or tooling.

Social Network Analysis (SNA) is applied to map intra-team and extra-team communication frequencies, information bottlenecks, and response latencies. This phase identifies how effectively live security telemetry dashboards serve as functional boundary objects to resolve organizational friction during an infrastructure crisis. Communication is captured across all available channels used during the simulation, synchronous voice, written incident-channel messages, and dashboard annotation events, and encoded into a time-stamped interaction graph for subsequent centrality and bottleneck analysis.

4.4 Validity, Reliability, and Ethical Considerations

To strengthen internal validity, all adversarial simulations in Phase II are executed against an identical baseline system configuration before any SOG-specific parameters are varied, isolating the causal contribution of the SOG itself. External validity is addressed by grounding the sandboxed enterprise environment in workflow patterns drawn from real, de-identified enterprise process documentation rather than purely synthetic task specifications. All simulated data used in Phases II and III is synthetic or fully de-identified; no production customer data, credentials, or proprietary enterprise systems are exposed to the adversarial evaluation engine at any stage of the research.

5 Anticipated Contributions, Discussion, and Industrial Impact

5.1 Theoretical and Academic Contributions

This dissertation provides a foundational contribution to the fields of Engineering Systems Management and Algorithmic Socio-Technical Governance.

- It expands Design Science literature by demonstrating how Dorst's co-evolution model can be computationally codified within automated system architectures, moving the model from a descriptive account of human designer cognition to an operational, machine-enforceable governance loop.
- It introduces a validated methodology for extending Zero-Trust principles from simple human identity verification to complex, non-deterministic agent intent verification, addressing a structural gap identified across the reviewed cybersecurity literature.
- It provides one of the first empirically grounded applications of Boundary Objects Theory to non-human, autonomous infrastructure, extending a body of theory previously developed almost exclusively in the context of human scientific and organizational collaboration.

5.2 Industrial and Practical Impact

For enterprise leaders and technology executives, this research delivers an actionable, implementation-ready toolkit. By deploying the Security Orchestration Gateway and utilizing the quantified efficiency frameworks defined herein, organizations can resolve the paralyzing gridlock between innovation velocity and risk mitigation.

This framework provides the empirical blueprint necessary to confidently scale autonomous Agentic AI infrastructures across highly regulated, safety-critical industries, including defense, healthcare, fintech, and critical national infrastructure, ensuring long-term systemic resilience in an era of automated threats. Beyond the technical artifact itself, the Phase III organizational findings offer compliance and human-resources functions a concrete, evidence-based argument for investing in cross-functional team composition specifically in security-critical operational roles, rather than treating team structure as an incidental staffing decision.

5.3 Discussion

The proposed framework’s central contribution is not any single one of its three theoretical layers, but the explicit articulation of how they depend on one another in practice. A purely technical reading of the problem, deploy a sufficiently sophisticated Zero-Trust gateway and consider the matter solved, would predict that Phase III’s organizational analysis is unnecessary. The framework’s co-evolutionary premise suggests otherwise: because the threat landscape continuously reshapes itself in response to agent capability changes, no static technical control, however sophisticated at the moment of deployment, remains sufficient without an organizational apparatus capable of continuously reinterpreting and updating it. This is the theoretical justification for treating the human, cross-functional coordination layer as a first-class component of the governance architecture rather than an auxiliary concern.

6 Limitations

Several limitations of the proposed research design should be acknowledged. First, the sandboxed evaluation environment in Phase II, while grounded in real enterprise workflow patterns, cannot fully replicate the scale, legacy-system heterogeneity, and organizational politics of production deployments; findings regarding defensive efficiency and throughput degradation should therefore be treated as indicative rather than directly generalizable without further field validation. Second, the adversarial evaluation engine, however sophisticated, is itself a designed artifact and may systematically underrepresent attack strategies that fall outside the assumptions of its designers, a limitation common to all red-team-style evaluation methodologies. Third, the Phase III case-study comparison of cross-functional versus homogeneous teams uses simulated incidents; team behavior under genuine, career-consequential incident conditions may differ from behavior observed in a research simulation, even when participants are experienced practitioners. Finally, the framework’s reliance on Common Criteria EAL 4+ as its compliance benchmark, while appropriate for many regulated contexts, may require adaptation for jurisdictions or industries governed by materially different regulatory regimes.

7 Future Work

Several extensions of this research program are anticipated. First, longitudinal field deployment of the SOG within a live, non-simulated enterprise environment would allow the quantitative metrics developed in Phase II to be validated against real operational conditions, including legacy-system heterogeneity and adversarial pressure that a sandboxed environment cannot fully replicate. Second, the behavioral baselining

component of the Algorithmic Zero-Trust layer could be extended to support federated learning across multiple enterprise deployments, allowing anomaly detection models to benefit from cross-organizational threat intelligence without requiring any single organization to expose its raw telemetry to another. Third, the Phase III organizational findings suggest a promising avenue for follow-on research into the specific team-composition variables, beyond the cross-functional versus homogeneous distinction examined here, that most strongly predict incident-response performance, including variables such as team tenure, prior incident experience, and the specific configuration of the boundary objects made available to the team. Finally, future work should examine how the framework generalizes to agentic systems operating with even greater autonomy, including systems capable of modifying their own tool access or provisioning new infrastructure without any human approval step in the loop.

8 Conclusion

This research set out to address a specific and increasingly urgent gap: the absence of a governance framework capable of matching the pace and non-determinism of autonomous, agentic AI infrastructure. By synthesizing Dorst’s problem-solution co-evolution model, an Algorithmic Zero-Trust extension of cybersecurity architecture, and Boundary Objects Theory from innovation management, this thesis develops a socio-technical framework, anchored by the Security Orchestration Gateway, that treats security, compliance, and organizational coordination as a single, co-evolving system rather than as separately optimized silos. The three-phase methodology, spanning artifact design, adversarial empirical validation, and organizational boundary-spanning analysis, is intended to produce not only a technically defensible security architecture but also an empirically grounded account of the human coordination structures required to sustain it. As enterprises across defense, healthcare, fintech, and critical infrastructure continue to expand the operational authority granted to autonomous agents, frameworks of this kind will be necessary to ensure that the resulting systems remain resilient, auditable, and aligned with the organizations they are meant to serve.

References

- [1] Chase, H. (2023). LangChain: Building applications with LLMs through composability and agentic workflows. GitHub.
- [2] Dorst, K. (2019). Co-evolution and the engine of innovation: Behavioral design spaces. *Design Studies*, 62, 55–73.
- [3] George, M. L. (2002). *Lean Six Sigma: Combining Six Sigma Quality with Lean Speed*. McGraw-Hill.
- [4] Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Rossow, C., & Fritz, M. (2023). Not what you’ve signed up for: Compromising Next-Gen LLM applications via indirect prompt injection. arXiv preprint arXiv:2302.12173.
- [5] Holling, C. S. (1973). Resilience and stability of ecological systems. *Annual Review of Ecology and Systematics*, 4(1), 1–23.
- [6] Jansson, D. G., & Smith, S. M. (1991). Design fixation. *Design Studies*, 12(1), 3–11.
- [7] Perrow, C. (1984). *Normal Accidents: Living with High-Risk Technologies*. Basic Books.
- [8] Rose, S., Borchert, O., Mitchell, S., & Connelly, S. (2020). Zero Trust Architecture. NIST Special Publication 800-207.

- [9] Star, S. L., & Griesemer, J. R. (1989). Institutional ecology, translations' and boundary objects: Amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907–39. *Social Studies of Science*, 19(3), 387–420.
- [10] Zou, A., Wang, Z., Kolter, J. Z., & Fredrikson, M. (2023). Universal and transferable adversarial attacks on aligned language models. arXiv preprint arXiv:2307.15043.

A Glossary of Terms

Agentic AI Autonomous, multi-agent systems capable of dynamic goal decomposition, independent tool-use, and execution of multi-step workflows with minimal human mediation.

Algorithmic Zero-Trust

An extension of classical Zero-Trust Architecture in which the unit of authentication is an agent's semantic intent rather than a static identity or service credential.

Boundary Object

An artifact, such as a compliance dashboard or cryptographic log, that is flexible enough to be locally meaningful to distinct organizational communities while maintaining a shared, common identity across them.

Cascading Algorithmic Failure

A failure mode in which small perturbations in one agent's output compound into systemic operational disruption across a network of interdependent agents.

Common Criteria EAL 4+

An internationally recognized information security certification standard (Evaluation Assurance Level 4, augmented) frequently required in regulated and safety-critical procurement contexts.

Defensive Efficiency (E_d)

The ratio of successfully contained adversarial exploits to the total number of injected exploits in an adversarial evaluation.

Indirect Prompt Injection

An attack in which malicious instructions are embedded within third-party content that an agent later ingests as ordinary data, rather than being submitted directly by an adversarial user.

Problem-Solution Co-Evolution

A design science model, originating with Dorst, in which the designer's understanding of a problem and the candidate solution space evolve jointly and iteratively rather than sequentially.

Security Orchestration Gateway (SOG)

The semantic abstraction layer proposed in this thesis that intercepts agent-to-agent and agent-to-system communication and evaluates it against a deterministic policy matrix.

Semantic Compiler

The SOG subcomponent that translates an agent's probabilistic reasoning trace and proposed tool invocation into a normalized, deterministic intermediate representation.

Social Network Analysis (SNA)

A methodology for mapping and quantifying communication frequencies, information bottlenecks, and response latencies within and across organizational teams.

Throughput Degradation Coefficient (D_t)

A metric evaluating the operational penalty of security friction, comparing gated transaction throughput against ideal baseline throughput.

Zero-Trust Architecture (ZTA)

A security model, formalized by NIST, operating on the principle of “never trust, always verify,” in which no user, device, or service is implicitly trusted regardless of network location.